

BIGRAM LANGUAGE MODELS

From Counting to a Neural Network

A study-notes report on building a character-level name generator, following Andrej Karpathy's makemore.

32,033

names in dataset

27

token vocabulary (a-z + .)

2.454

avg negative log-likelihood

228,146

bigram transitions counted

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Repository: BigramModels

Generated 2026

Project Overview

SECTION 01

This project re-implements a character-level language model that generates new, name-like words. The model learns the statistics of which letter tends to follow which, then samples fresh names one character at a time.

Two complementary approaches are explored side by side:

1. A statistical bigram model that simply counts how often each pair of adjacent characters occurs, then normalises those counts into probabilities.
2. A single-layer neural network that learns an equivalent probability table through gradient-based optimisation, framing the exact same problem as machine learning.

Both are evaluated with the same yardstick — the average negative log-likelihood (NLL) — so the two worlds can be compared directly. The dataset is the classic list of ~32k English first names used in Karpathy's makemore series.

Names dataset → **Count bigrams** → **Probability matrix P** → **Sample new names** → **Evaluate NLL loss**

End-to-end pipeline

First names in the dataset

emma olivia ava isabella sophia charlotte mia

Each name is wrapped with a special '.' boundary token marking where a word starts and ends.

The Bigram Count Matrix

SECTION 02

Every cell counts how often the row-character is immediately followed by the column-character across all 32k names. Brighter = more frequent. This 27x27 table is the entire 'knowledge' of the statistical model.

Bigram frequency heatmap (rows = current char, cols = next char)

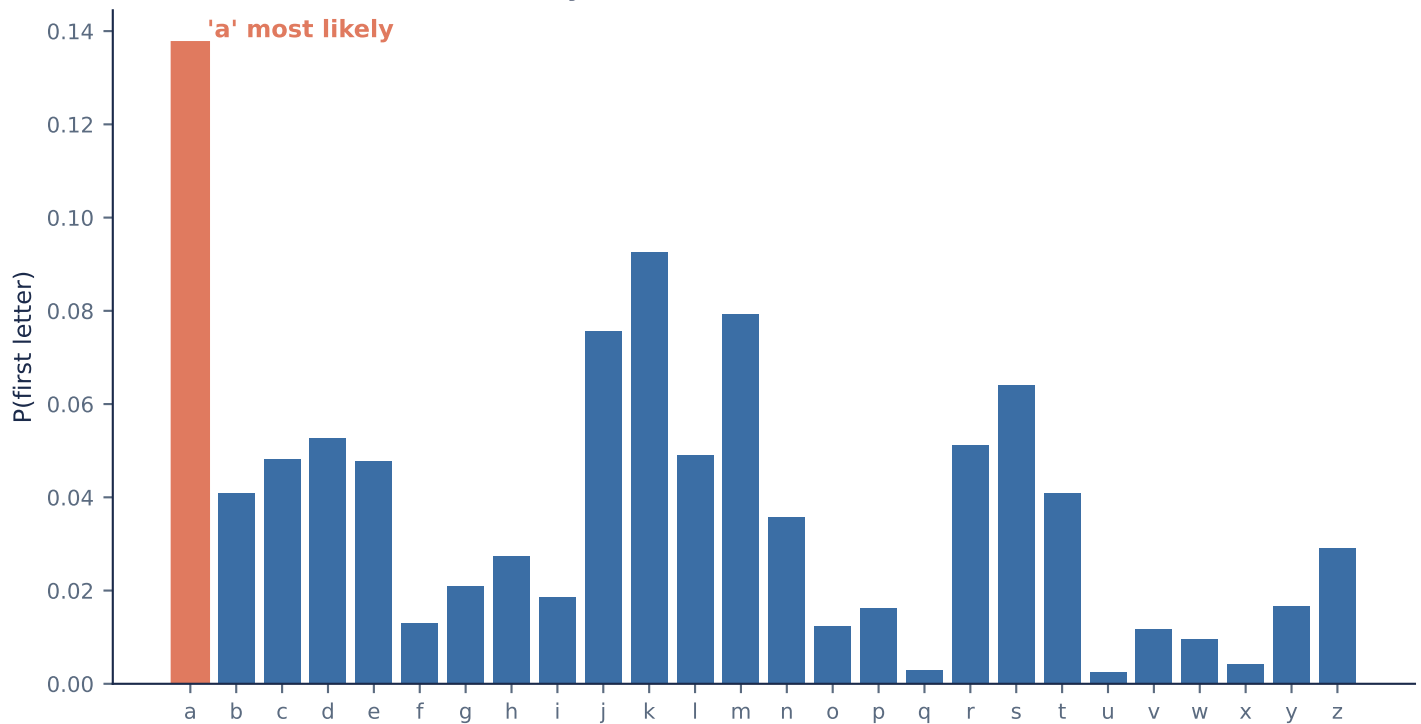
.0	a.432	b.1306	c.1542	d.1690	e.1531	f.417	g.669	h.874	i.591	j.2422	k.2963	l.1572	m.2538	n.1146	o.394	p.515	q.92	r.1639	s.2055	t.1308	u.78	v.376	w.307	x.134	y.535	z.929
a.6640	aa.556	ab.541	ac.470	ad.1042	ae.692	af.134	ag.168	ah.2332	ai.1650	aj.175	ak.568	al.2528	am.1634	an.5438	ao.63	ap.82	aq.60	ar.3264	as.1118	at.687	au.381	av.834	aw.161	ax.182	ay.2050	az.435
b.114	ba.321	bb.38	bc.1	bd.65	be.655	bf.0	bg.0	bh.41	bi.217	bj.1	bk.0	bl.103	bm.0	bn.4	bo.105	bp.0	bq.0	br.842	bs.8	bt.2	bu.45	bv.0	bw.0	bx.0	by.83	bz.0
c.97	ca.815	cb.0	cc.42	cd.1	ce.551	cf.0	cg.2	ch.664	ci.271	cj.3	ck.316	cl.116	cm.0	cn.0	co.380	cp.1	cq.11	cr.76	cs.5	ct.35	cu.35	cv.0	cw.0	cx.3	cy.104	cz.4
d.516	da.1303	db.1	dc.3	dd.149	de.1283	df.5	dg.25	dh.118	di.674	dj.9	dk.3	dl.60	dm.30	dn.31	do.378	dp.0	dq.1	dr.424	ds.29	dt.4	du.92	dv.17	dw.23	dx.0	dy.317	dz.1
e.3983	ea.679	eb.121	ec.153	ed.384	ee.1271	ef.82	eg.125	eh.152	ei.818	ej.55	ek.178	el.3248	em.769	en.2675	eo.269	ep.83	eq.14	er.1958	es.861	et.580	eu.69	ev.463	ew.50	ex.132	ey.1070	ez.181
f.80	fa.242	fb.0	fc.0	fd.0	fe.123	ff.44	fg.1	fh.1	fi.160	fj.0	fk.2	fl.20	fm.0	fn.4	fo.60	fp.0	fq.0	fr.114	fs.6	ft.18	fu.10	fv.0	fw.4	fx.0	fy.14	fz.2
g.108	ga.330	gb.3	gc.0	gd.19	ge.334	gf.1	gg.25	gh.360	gi.190	gj.3	gk.0	gl.32	gm.6	gn.27	go.83	gp.0	gq.0	gr.201	gs.30	gt.31	gu.85	gv.1	gw.26	gx.0	gy.31	gz.1
h.2409	ha.2244	hb.8	hc.2	hd.24	he.674	hf.2	hg.2	hh.1	hi.729	hj.9	hk.29	hl.185	hm.117	hn.138	ho.287	hp.1	hq.1	hr.204	hs.31	ht.71	hu.166	hv.39	hw.10	hx.0	hy.213	hz.20
i.2489	ia.2445	ib.110	ic.509	id.440	ie.1653	if.101	ig.428	ih.95	ii.82	ij.76	ik.445	il.1345	im.427	in.2126	io.588	ip.53	iq.52	ir.849	is.1316	it.541	iu.109	iv.269	iw.8	ix.89	iy.779	iz.277
j.71	ja.1473	jb.1	jc.4	jd.4	je.440	jf.0	jj.0	jh.45	ji.119	jj.2	jk.2	jl.9	jm.5	jn.2	jo.479	jp.1	jq.0	jr.11	js.7	jt.2	ju.202	jv.5	jw.6	jx.0	jy.10	jz.0
k.363	ka.1731	kb.2	kc.2	kd.2	ke.895	kf.1	kg.0	kh.307	ki.509	kj.2	kk.20	kl.139	km.9	kn.26	ko.344	kp.0	kq.0	kr.109	ks.95	kt.17	ku.50	kv.2	kx.0	ky.379	kz.2	
l.1314	la.2623	lb.52	lc.25	ld.138	le.2921	lf.22	lg.6	lh.19	li.2480	lj.6	lk.24	ll.1345	lm.60	ln.14	lo.692	lp.15	lq.3	lr.18	ls.94	lt.77	lu.324	lv.72	lw.16	lx.0	ly.1588	lz.10
m.516	ma.2590	mb.112	mc.51	md.24	me.818	mf.1	mg.0	mh.5	mi.1256	mj.7	mk.1	ml.5	mm.168	mn.20	mo.452	mp.38	mq.0	mr.97	ms.35	mt.4	mu.139	mv.3	mw.2	mx.0	my.287	mz.11
n.6763	na.2977	nb.8	nc.213	nd.704	ne.1359	nf.11	ng.273	nh.26	ni.1725	nj.44	nk.58	nl.195	nm.19	nn.1906	no.496	np.5	nq.2	nr.44	ns.278	nt.443	nu.96	nv.55	nw.11	nx.6	ny.465	nz.145
o.855	oa.149	ob.140	oc.114	od.190	oe.132	of.34	og.44	oh.171	oi.69	oj.16	ok.68	ol.619	om.261	on.2411	oo.115	op.95	oq.3	or.1059	os.504	ot.118	ou.275	ov.176	ow.114	ox.45	oy.103	oz.54
p.33	pa.209	pb.2	pc.1	pd.0	pe.197	pf.1	pg.0	ph.204	pi.61	pj.1	pk.1	pl.16	pm.1	pn.1	po.59	pp.39	pq.0	pr.151	ps.16	pt.17	pu.4	pv.0	pw.0	px.0	py.12	pz.0
q.28	qa.13	qb.0	qc.0	qd.0	qe.1	qf.0	qg.0	qh.0	qi.13	qj.0	qk.0	ql.1	qm.2	qn.0	qo.2	qp.0	qq.0	qr.1	qs.2	qt.0	qu.206	qv.0	qw.3	qx.0	qy.0	qz.0
r.1377	ra.2356	rb.41	rc.99	rd.187	re.1697	rf.9	rg.76	rh.121	ri.3033	rj.25	rk.90	rl.413	rm.162	rn.140	ro.869	rp.14	rq.16	rr.425	rs.190	rt.208	ru.252	rv.80	rw.21	rx.3	ry.773	rz.23
s.1169	sa.1201	sb.21	sc.60	sd.9	se.884	sf.2	sg.2	sh.1285	si.684	sj.2	sk.82	sl.279	sm.90	sn.24	so.531	sp.51	sq.1	sr.55	ss.461	st.765	su.185	sv.14	sw.24	sx.0	sy.215	sz.10
t.483	ta.1027	tb.1	tc.17	td.0	te.716	tf.2	tg.2	th.647	ti.532	tj.3	tk.0	tl.134	tm.4	tn.22	to.667	tp.0	tq.0	tr.352	ts.35	tt.374	tu.78	tv.15	tw.11	tx.2	ty.341	tz.105
u.155	ua.163	ub.103	uc.103	ud.136	ue.169	uf.19	ug.47	uh.58	ui.121	uj.14	uk.93	ul.301	um.154	un.275	uo.10	up.16	uq.10	ur.414	us.474	ut.82	uu.3	uv.37	uw.86	ux.34	uy.13	uz.45
v.88	va.642	vb.1	vc.0	vd.1	ve.568	vf.0	vg.0	vh.1	vi.911	vj.0	vk.3	vl.14	vm.0	vn.8	vo.153	vp.0	vq.0	vr.48	vs.0	vt.0	vu.7	vv.7	vw.0	vx.0	vy.121	vz.0
w.51	wa.280	wb.1	wc.0	wd.8	we.149	wf.2	wg.1	wh.23	wi.148	wj.0	wk.6	wl.13	wm.2	wn.58	wo.36	wp.0	wq.0	wr.22	ws.20	wt.8	wu.25	wv.0	ww.2	wx.0	wy.73	wz.1
x.164	xa.103	xb.1	xc.4	xd.5	xe.36	xf.3	xg.0	xh.1	xi.102	xj.0	xk.0	xl.39	xm.1	xn.1	xo.41	xp.0	xq.0	xr.0	xs.31	xt.70	xu.5	xv.0	xw.3	xx.38	xy.30	xz.19
y.2007	ya.2143	yb.27	yc.115	yd.272	ye.301	yf.12	yg.30	yh.22	yi.192	yj.23	yk.86	yl.1104	ym.148	yn.1826	yo.271	yp.15	yq.6	yr.291	ys.401	yt.104	yu.141	yv.106	yw.4	yx.28	yy.23	yz.78
z.160	za.860	zb.4	zc.2	zd.2	ze.373	zf.0	zg.1	zh.43	zi.364	zj.2	zk.2	zl.123	zm.35	zn.4	zo.110	zp.2	zq.0	zr.32	zs.4	zt.4	zu.73	zv.2	zw.3	zx.1	zy.147	zz.45

Reading the Statistics

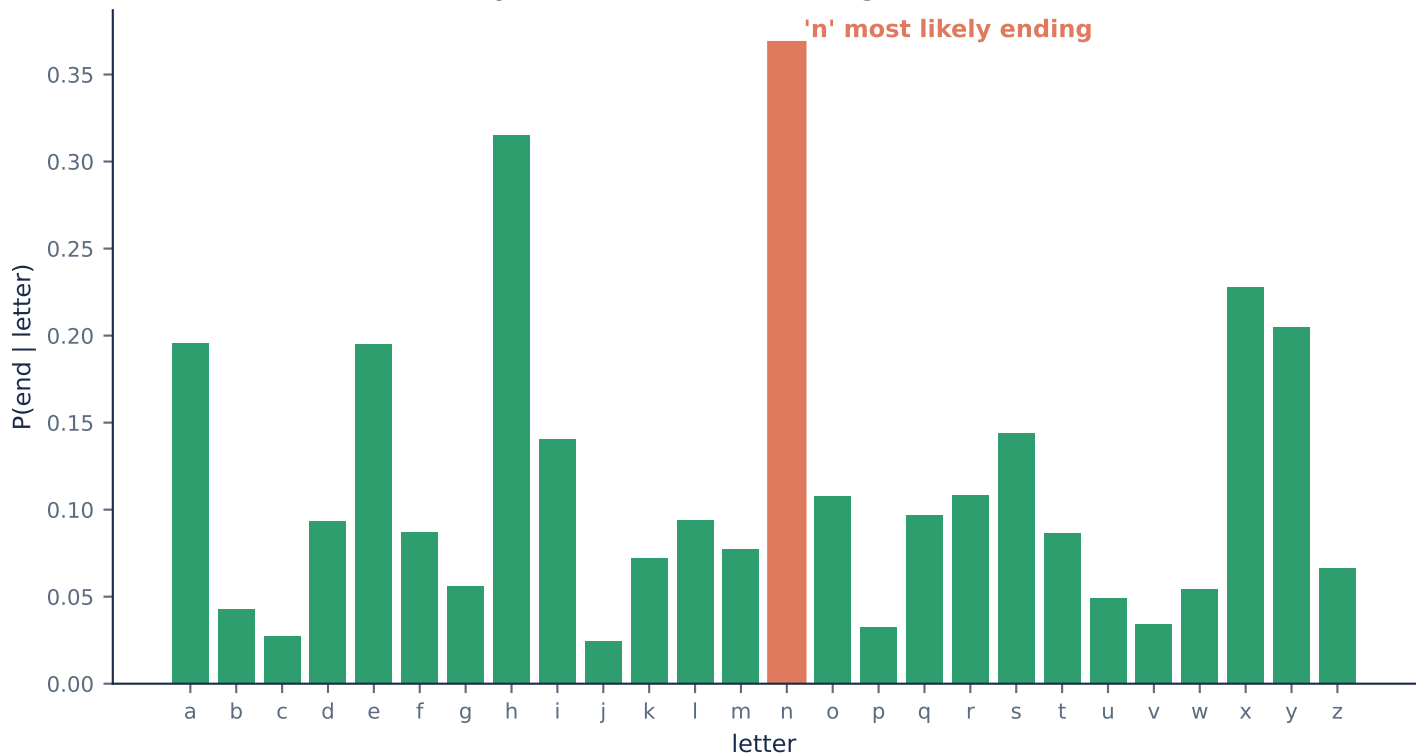
SECTION 03

The probability matrix P is just the count matrix normalised row-by-row (with +1 smoothing). Slicing it reveals interpretable structure:

Probability of each letter STARTING a name



Probability that a name ENDS right after each letter



Names invented by the model

Starting from '.', repeatedly sample the next character from P until '.' is drawn again:

- cexze
- momasurailezitynn
- konimittain
- llayn
- ka
- da
- staiyabrthrigotai
- moliellavo

Not real names, but clearly name-like — the model has captured plausible letter transitions.

How good is the model? Negative Log-Likelihood (NLL)

The quality of the model is measured by the average negative log-likelihood over every bigram in the dataset. A perfect model scores 0; lower is better. Equivalently, $\text{loss} = -\text{mean}(\log P(\text{next} \mid \text{current}))$.

2.4544

average NLL (whole dataset)

+1

model smoothing

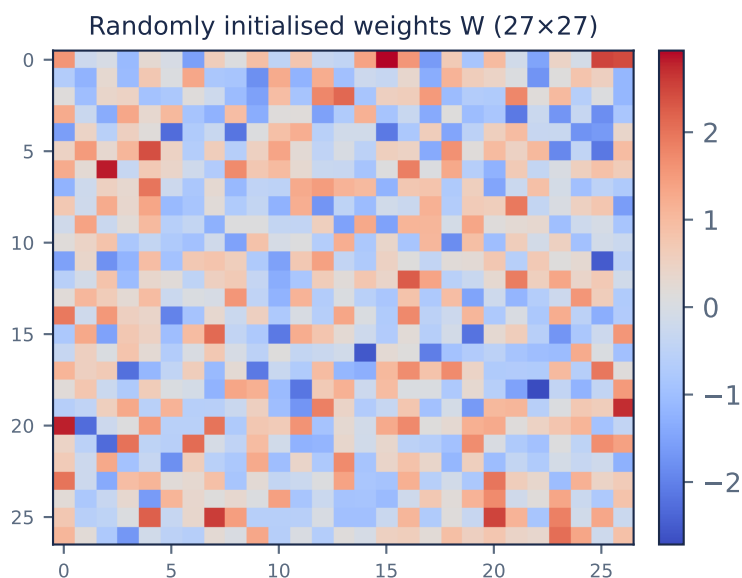
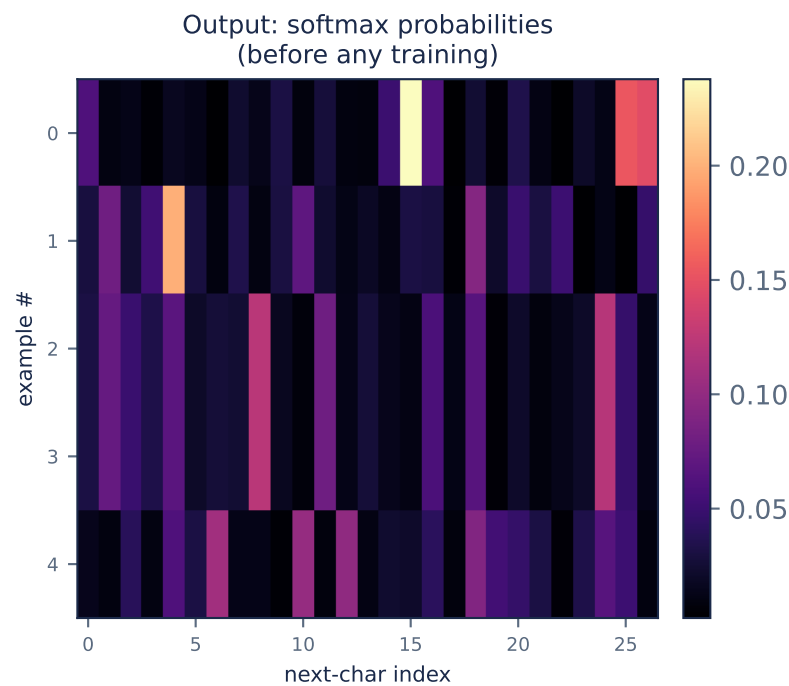
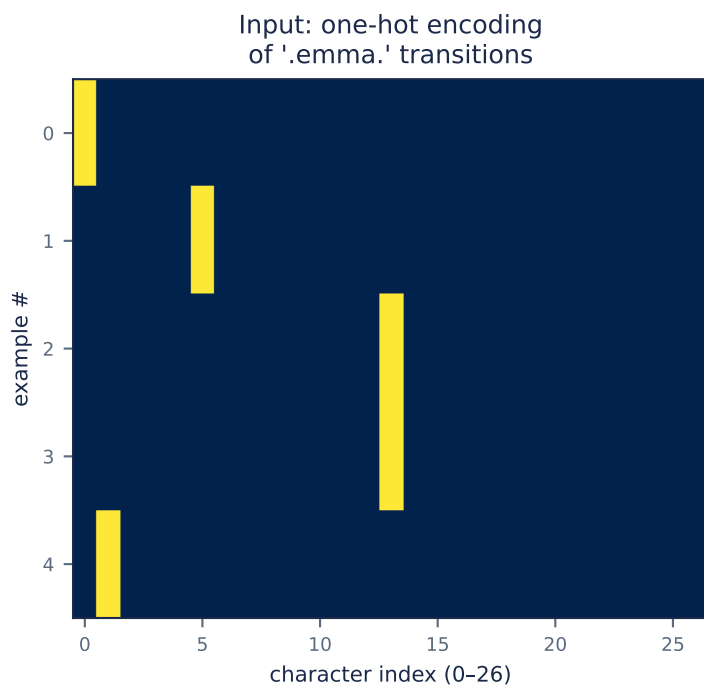
Laplace smoothing added to every count so no transition has zero probability — keeping the loss finite even for unseen pairs like 'jq'.

The Neural Network View

SECTION 05

The same problem, reframed as learning. Instead of counting, a single linear layer (27→27 weights) turns a one-hot encoded input character into a probability distribution over the next character via the softmax function.

```
one-hot(x) → logits = x · W → counts = exp(logits) → probs = counts / Σcounts
```



Initial loss (untrained)

3.7693

Gradient descent on W will drive this down toward the statistical model's NLL.

1. Two routes, one answer

Counting bigrams and training a 27×27 linear layer arrive at the same probability table — one by direct statistics, one by gradient descent. Seeing this equivalence is the core lesson.

2. Smoothing matters

Adding +1 to every count guarantees non-zero probabilities, so the negative log-likelihood never blows up on unseen transitions.

3. NLL is the compass

The fully-counted model reaches an average NLL of 2.4544 across the dataset — the baseline any learned model is measured against.

4. Bigram is the ceiling

Looking only one character back limits quality. The natural next step is to widen the context window (trigrams, MLPs, and eventually a Transformer) — the trajectory of the makemore series.

Reference: Andrej Karpathy — "The spelled-out intro to language modeling: building makemore".